HLApp User Guide

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Use this guide to prepare HLA genotypes, select outputs, interpret peptide-binding promiscuity and divergence, and export results.

The app supports HLA-A, HLA-B and HLA-C, HLA-DP, HLA-DQ and HLA-DRB1. The metrics describe different properties; a high score does not by itself establish clinical benefit.

Choose the output that matches your question

Metric	What it describes	Output level
Promiscuity	Breadth of the peptide sequence preferences of one HLA molecule	Individual alleles, means within HLA-A/B/C/DP/DQ/DRB1 and across HLA-I and II
HED	Physicochemical divergence between homologous HLA protein sequences	Separate values for HLA-A/B/C/DP/DQ/DRB1 and means across HLA-I and II
Functional divergence	Complementarity of peptide-binding submotifs for an allele pair	Separate values for HLA-A/B/C/DP/DQ/DRB1 and means across HLA-I and II
Homozygosity	Whether only one distinct allele remains at a gene	Gene-specific flags

Quick start

1. Download example data on the Calculate HLA metrics tab and prepare one row per sample with a unique ID.

2. Upload the table, choose the ID column and one representation of the HLA genotype, and set the delimiter used inside allele-list cells.

3. Select the metrics to calculate: allele or locus promiscuity, homozygosity, HED and/or functional divergence. Choose Get results, review completeness, then export.

Navigation: input preparation on page 2; calculation workflow on page 3; promiscuity on page 4; HED on page 5; functional divergence on page 6; additional tools on page 7; troubleshooting and verification on page 8; references on page 9.

Prepare the input table

Use a rectangular table with column names in the first row and one sample per subsequent row. Include a unique, nonempty sample ID. Column names are selected in the interface and need not follow a fixed naming scheme. Remove titles, notes, merged cells and extra header rows.

File	Confirmed reader behavior
CSV	Comma-separated columns. Quote any cell that itself contains commas.
TSV	Tab-separated columns.
XLSX or XLS	Excel table; the reader uses its default first worksheet.
TXT	Comma-separated by default in the upload module. Prefer CSV or TSV to avoid ambiguity.

Two ways to represent genotypes

1. Separate columns. Store the two copies of each gene in separate columns, including both identical copies for a known homozygote. For example:

```
ID,A1,A2,B1,B2,C1,C2
S01,A*01:01,A*03:01,B*51:01,B*57:01,C*07:01,C*15:02
S02,A*02:01,A*02:01,B*13:02,B*18:01,C*06:02,C*07:01
```

2. Allele lists. Store multiple alleles in a cell, separated by commas, semicolons or spaces; select the matching delimiter in step 2. The outer file delimiter and the delimiter inside a cell are different settings. Comma is the recommended internal delimiter.

```
ID,HLA_I
S01,"A0101,A0301,B5101,B5701,C0701,C1502"
```

The app example contains both HLA_I/HLA_II lists and separate gene columns. Select either the lists or the separate columns for a sample, not both. Combining duplicate representations can inflate allele counts and prevent calculation.

Allele names and class II complexes

Use uppercase, two-field allele names with the locus included: A*01:01 or A0101; DRB1*07:01 or DRB10701. The importer removes several punctuation characters and truncates longer numeric names to four digits. Use canonical compact names if detection fails. Do not assume that low-resolution typing, ambiguous calls, null-expression suffixes or every modern allele name is supported.

```
DP example: DPA10103-DPB10301
DQ example: DQA10201-DQB10303
DR example: DRB10701
```

For DP and DQ, provide explicit alpha-beta complexes, or separate alpha- and beta-chain calls. With separate calls, the code forms every alpha-beta combination within the sample, including potential trans combinations, and discards DQ combinations if they are unable to

form stable complexes (i.e., they are forbidden pairings [1]). DR is represented by DRB1 alone.

Known homozygosity is different from incomplete typing. A single allele in a list does not prove homozygosity. Preserve both copies where known; never fill an unknown second allele by copying the first.

Calculate and export results

Step 1 Upload file

Open “Calculate HLA metrics” and remain on the “1. Upload file” tab.
Use Browse... to choose your table. Download example data provides the app’s own CSV template.
After the file is read successfully, select Next ->.
If the table cannot be read, check the file format and remove any non-table lines.

Step 2 Select input columns

Choose the column of sample IDs, then tick the HLA columns to analyze. Select / Deselect all is convenient only when every detected column is needed. For example, in the published example data, it would select both alternative genotype representations, so you have to choose columns individually.

Set the internal delimiter to comma, space or semicolon. Inspect Your input, which previews the first three rows. The source checks for candidate HLA columns in the first six rows and checks whether alleles can be recognized within the first 100 rows. A successful recognition message does not validate every sample. Ensure all intended loci are selected, then click Next ->.

Step 3 Choose output

Control	Selection and meaning
Allele promiscuity	Select loci for allele-level promiscuity values
Locus promiscuity	Select loci for calculating the mean of available scores within each locus.
Homozygosity	Choose yes to include gene-specific homozygosity flags.
HLA evolutionary divergence	Choose HLA-I and/or HLA-II for class-level HED values.
Functional divergence	Select loci for calculating functional divergence between alleles within each locus

The output choices are updated from recognized loci. Select at least one output and resolve excessive-allele errors (i.e., when more than two alleles are provided for a given locus) to enable Get results. Go back returns to the preceding step. Get results calculates the selected summaries.

Read and export the table

The results are returned in a table format with all the selected calculated metrics for each ID in one row. You can export the results in .xlsx and .csv format or copy them to clipboard.

You can select any functional divergence value in the table and click on the “Visualize functional divergence” button to see and compare submotifs of peptides bound by the two alleles at the given locus.

HLA peptide binding promiscuity

Promiscuity describes how broadly an HLA molecule can bind peptide sequences.

Calculating promiscuity for HLA class I molecules

In the original publication, we analyzed positive IEDB binding or T-cell assay records for 8–12-residue peptides, retaining alleles with at least 400 distinct peptides [2]. In the current implementation of the app, promiscuity was calculated using the Sarkizova et al. immunopeptidomics dataset [3] to avoid bias caused by heterogeneous study results deposited in the IEDB. For each peptide length L and position y, amino-acid frequencies P are compared with human-proteome background frequencies Q by calculating the Kullback-Laibler divergence value. [2]

$$D_{KL,L}(y) = \Sigma_a P_L(a,y) \log[P_L(a,y) / Q(a)]$$

A value of 10^{-7} is added to all amino-acid frequencies at a position containing a zero frequency. Position values are averaged within each length and corrected by subtracting the mean from 1,000 random human-proteome peptide samples of matching size and length. Length-specific corrected values are weighted by their repertoire proportions. Promiscuity is the reciprocal. [2]

$$D_{corrected,L} = (1/L)\Sigma_y D_{KL,L}(y) - D_{random,L}$$

$$Pr = 1 / [\Sigma_L w_L D_{corrected,L}]$$

Here w_L is the fraction of peptides of length L. Lower corrected divergence from background gives higher Pr. [2]

Calculating promiscuity for HLA class II molecules

To determine promiscuity for HLA-II molecules, we applied the same approach to unique nine-residue binding cores from a large immunopeptidomics dataset [4]. We retained alleles with at least 100 core sequences.

App implementation

The app calculates promiscuity for each selected locus. Locus promiscuity is an unweighted arithmetic mean of alleles for HLA-A/B/C/DRB1 and alpha-beta chain combinations for HLA-DP/DQ. Missing scores are ignored during calculation.

$$Pr_{mean} = (1/k)\Sigma_j Pr_j$$

Here k is the number of available scores at the locus.

HLA evolutionary divergence

HED quantifies physicochemical differences between the two homologous HLA protein sequences. The Grantham amino-acid distance reflects composition, polarity and volume. For aligned peptide-binding domains of length N, HED is the mean site-wise Grantham distance. Identical sequences give zero. According to Chowell et al., we averaged the HLA-A, HLA-B and HLA-C pairwise values to obtain mean HLA-I HED. [5–7]

$$HED(u,v) = (1/N)\Sigma_i G(u_i, v_i)$$

$$HED_I = [HED_A + HED_B + HED_C] / 3$$

G is the Grantham distance for the two amino acids at aligned position i. Higher HED denotes more physicochemical sequence divergence; it is a proxy for binding-repertoire differences rather than a direct measurement of peptide overlap. [5,6]

For HLA-II, distances are compared within DPA1, DPB1, DQA1, DQB1 and DRB1, then the five values are averaged.

$$HED_{II} = [HED_{DPA1} + HED_{DPB1} + HED_{DQA1} + HED_{DQB1} + HED_{DRB1}] / 5$$

HED values for allele pairs at the same locus are calculated using the perl script published by Pierini et al. [6, https://sourceforge.net/projects/granthamdist/]. HLA allele sequences were downloaded from the same resource.

HED values for HLA-I and HLA-II are only calculated if all loci are represented and have valid HED values.

HLA functional divergence

HLA functional divergence describes the degree to which an individual's HLA allele at given locus differ in their peptide-binding preferences, reflecting their capacity to present distinct sets of antigens to T cells.

Calculating functional divergence

Following the approach by Viard et al., we determined nonamer peptide clusters from 95 HLA-I alleles. We summarized clusters as submotifs, retaining highly selective position–amino-acid features with an entropy value below 0.1. We then compared the submotif overlap of two alleles yielding a divergence measure: greater divergence means lower overlap between the repertoire of the two alleles. As in the original study, singletons were retained as a separate subgroup. [8]

We applied the same approach for HLA-II alleles by using unique nine-residue binding cores from a large immunopeptidomics dataset [ref].

Warning. The original publication's calculation scripts were not available to the app developers. While the reproduced values show strong correlation with the ones in the original publication (Spearman's rho was 0.84, 0.79, and 0.9 for HLA-A, B, and C, respectively), some of them may differ. Therefore, the numeric result should be reported as an app-derived reproduction, with its version stated.

Explore motifs and translate allele names

Submotif explorer

This module visualizes peptide clusters and submotifs used for calculating functional divergence between HLA alleles (see page 6. and ref. [8]).

Choose “Submotif explorer”, then HLA-I or HLA-II under “Select HLA class”
When the data are loaded, “Select allele form list, or start typing”: chooses a molecule
Select a cluster: chooses the peptide cluster of the selected allele (the clustered are ordered by decreasing number of peptides).
The general view shows the molecule’s motif, total peptide count and an NMDS plot; the selected cluster is highlighted with green.

On the right side, you can see the submotif of the selected peptide cluster, the peptide count, and the entropy of amino acids at each position. On the entropy plot, the red dashed line indicates the entropy value of 0.1, which is used for defining the submotif (see page 6.).

Peptide sequences can be visualized with the “Show peptides/Hide peptides” button. All data on the submotif can be downloaded:

Download control	File and intended use
Download Submotif Plot	SVG vector graphic
Download Submotif Data	RDS motif data for analysis in R
Download Peptide List	TXT peptide list
Download Entropy Plot	PNG graphic
Download Entropy Data	RDS vector of entropy values

Submotif comparison

Choose “Submotif comparison” and select under Allele 1 and Allele 2. The code displays each molecule’s binding motif and the submotif composition text, and also the functional-divergence value for the two alleles.

Translate allele formats

Enter canonical compact identifiers, one per line, in the “Enter alleles” field; for example A0101, DRB10701 or DQA10201-DQB10303. Alternatively, select the “Load alleles from PromsicuityApp” to import unique allele names in the “Calculate HLA metrics” module. Select NetMHCpan, MHCnuggets and/or MixMHCpred.

The output includes columns for NetMHCpan/NetMHCIIpan, MHCnuggets, and MixMHCpred/MixMHC2pred as selected. Unsupported entries trigger a warning and are excluded from the translation table.

Troubleshoot and verify the analysis

Symptom	Action
No candidate input ID column	Use unique, nonempty IDs in a dedicated column that contains no HLA allele calls.
No HLA columns or alleles recognized	Use uppercase locus-prefixed two-field calls. Put valid calls near the top; check the internal delimiter and selected columns.
More than two alleles error	Select one genotype representation. Remove accidental duplicate columns. The code permits up to four DP or DQ complexes but only two A, B, C or DRB1 entries.
Only one allele warning	Check that both gene copies are present where required. Do not replace unknown calls with fabricated homozygosity.
Blank NA or NaN result	Check allele or pair coverage. Missing lookup values are not evidence of zero. Available-case averages can hide incomplete coverage.
Empty plots or disabled results	Check input selections and data availability. Reload the app if the session stops responding.

Check every sample before interpretation

Check exported IDs against the input, review each required gene and score, and count missing outputs. Preserve known homozygous copies, record excluded DQ combinations, and state whether partial-case means were accepted. Save the input, selected columns and delimiters, output choices, export and analysis date. These records make comparisons reproducible.

Scientific and implementation references

Bracketed numbers refer to these sources. Links open the original methods or implementation. Access date for web and repository material is 13 October 2026.

[1] Raymond CK et al. Ancient haplotypes of the HLA Class II region. Genome Research. 2005;15.9:1250-1257. Definition of forbidden HLA-DQ alpha and beta chain pairings.

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[2] Manczinger M et al. Negative trade-off between neoantigen repertoire breadth and the specificity of HLA-I molecules shapes antitumor immunity. Nature Cancer. 2021;2:950–961. Original HLA-I KL-based promiscuity method.

doi.org/10.1038/s43018-021-00226-4

[3] Sarkizova S et al. A large peptidome dataset improves HLA class I epitope prediction across most of the human population. Nature Biotechnology. 2020;38:199–209. HLAthena peptide data underlying the published submotif approach.

doi.org/10.1038/s41587-019-0322-9

[4] HLA-II peptide-binding diversity shapes humoral immune responses and susceptibility to infections. Nature Communications. 2026. Methods section Determining the promiscuity of HLA-II molecules.

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[5] Chowell D et al. Evolutionary divergence of HLA class I genotype impacts efficacy of cancer immunotherapy. Nature Medicine. 2019;25:1715–1720. HED calculation and HLA-I aggregation.

doi.org/10.1038/s41591-019-0639-4

[6] Pierini F and Lenz TL. Divergent Allele Advantage at Human MHC Genes Signatures of Past and Ongoing Selection. Molecular Biology and Evolution. 2018;35:2145–2158. Sequence divergence and peptide repertoire analysis.

doi.org/10.1093/molbev/msy116

[7] Grantham R. Amino acid difference formula to help explain protein evolution. Science. 1974;185:862–864. Original physicochemical amino-acid distance.

doi.org/10.1126/science.185.4154.862

[8] Viard M et al. Impact of HLA class I functional divergence on HIV control. Science. 2024;383:319–325. Submotif-based method; supplementary Data S1 supplies reference pairwise values.

doi.org/10.1126/science.adk0777

Support

For app errors or feedback, use the promiscuity_app repository. Include the app date, selected settings and a minimal example without identifiable sample information (screenshot is preferred).